

TA-002 | Dataset Understanding

Dataset identity

Project: Mobile Phone Price Category Prediction Using Random Forest

Source file: mobile_dataset.csv

Purpose: understand the dataset before cleaning and modeling.

Dataset dimensions

Records: 1,019

Columns: 15

Complete price values: 1,019

Duplicate record groups: 2

Main finding

The file is a raw product-listing dataset. Numeric values are mixed with text specifications, so feature extraction is required before Random Forest training.

The dataset has a numeric price column but no Price_Category target column yet.

Column Structure and Data Types

Observed type

CSV parsing initially reads all 15 fields as text. Numeric fields can be converted; embedded specification strings need parsing.

Column inventory

Name - product model name; text identifier.

Spec Score - overall specification score; numeric text.

rating - product rating; numeric text.

price - listed price in rupees; numeric text and candidate source for categories.

img - product image URL; normally exclude from tabular modeling.

tag - LAUNCHED, UPCOMING, or RUMORED status; categorical text.

sim - SIM and connectivity details; text requiring extraction.

processor - processor, cores, and speed; text requiring extraction.

storage - RAM and internal storage together; split into numeric features.

battery - capacity and charging details; extract numeric capacity/power.

display - size, resolution, refresh rate, and design; extract relevant features.

camera - rear and front camera details; extract megapixels.

memoryExternal - memory-card information; optional categorical feature.

version - operating-system version; categorical/text feature.

fm - FM radio availability; optional categorical feature.

Data Quality Snapshot

Missing values

fm: 725 missing (71.1%)

memoryExternal: 351 missing (34.4%)

version: 48 missing (4.7%)

processor: 26 missing (2.6%)

storage: 20 missing (2.0%)

camera: 11 missing (1.1%)

battery: 6 missing (0.6%)

display: 1 missing (0.1%)

All other columns have no missing values.

Duplicates

Oppo Reno 14F, price 34,990, storage 8 GB RAM, 256 GB inbuilt: 2 occurrences

Samsung Galaxy M37 5G, price 17,990, storage 8 GB RAM, 128 GB inbuilt: 2 occurrences

Numeric validation

Numeric price values: 1,019 of 1,019.

Numeric and Categorical Summary

Numeric ranges

Spec Score: minimum 7, maximum 96.

Price: minimum Rs. 99, maximum Rs. 489,990, mean Rs. 35,006.71.

Descriptive price bands

Below 15,000: 288 records

15,000-29,999: 369 records

30,000 and above: 362 records

Categorical observations

Status: 664 LAUNCHED, 253 UPCOMING, 102 RUMORED.

FM: 294 explicitly marked No FM Radio; 725 blank.

Conclusions and Next Steps

Findings

1. Create a documented Price_Category rule from price before classification.
2. Parse RAM, storage, battery, display, camera, and processor text into numeric features.
3. Treat optional missing fields explicitly, especially fm and memoryExternal.
4. Review and remove duplicate records before train-test splitting.
5. Exclude img from a tabular model unless image modeling is specifically required.

Conclusion

mobile_dataset.csv contains 1,019 mobile-phone listings and 15 columns. It is suitable as a raw source after cleaning, feature extraction, duplicate handling, and creation of a documented price-category target.